

Local AI Agent With RAG

A **local-first RAG system** using **Ollama** (Llama 3.2) for inference and a custom vector retriever · no cloud APIs required.

· Features

- **Fully Local** · Runs entirely on your machine via Ollama
 - **RAG Pipeline** · Vector-based retrieval before answering
 - **Domain-Specific** · Pre-loaded with pizza restaurant review data
 - **Conversational** · Interactive Q&A with context injection
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.. Architecture

User Query · Vector Retriever · Relevant Reviews · LLM Prompt · Ollama (Llama 3.2) · Answer

· Getting Started

Prerequisites

- Python 3.10+, Ollama running, `ollama pull llama3.2`

```
cd LocalAIAgentWithRAG
pip install -r requirements.txt
python main.py
```

· Logic Flow

- 1 **Startup** · Vector store initialized with review data
 - 2 **Query** · User asks a question
 - 3 **Retrieval** · Most relevant reviews found via vector search
 - 4 **Context** · Reviews injected into LLM prompt template
 - 5 **Inference** · Ollama runs Llama 3.2 locally
 - 6 **Response** · Context-aware answer displayed
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· Dependencies

langchain-ollama, langchain, python-dotenv

· License

Educational project · use freely for learning and reference.

